ASSIGNMENT - I REPORT

Air Quality Monitoring Using Official & Community Sensors

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List of abbreviations

$PM_{2.5}$: Particulate Matter 2.5 micrometer

PM_{10}: Particulate Matter 10 micrometer

EEA: European Environment Agency

RIVM: National Institute for Public Health and the Environment (‘*Rijksinstituut voor Volksgezondheid en Milieu’* in dutch)

GGD: Medicine and Health Service (‘*Geneeskundige en Gezondheidsdienst’* in dutch)

DCMR: Service Central Environmental Management Rijnmond

EPA: Environmental Protection Agency

EC: European Commission

WHO: World Health Organisation

EU: European Union

NOx: Nitrogen Oxides

SOx: Sulphur Oxides

API: Application Programming Interface

CRS: Coordinate Reference System

OSM: OpenStreetMap

RMSE: Root Mean Square Error

1. Introduction

Air pollution poses significant risks to human health and the environment, contributing to respiratory and cardiovascular diseases, reducing quality of life, and leading to preventable deaths (U.S. EPA, 2024). Particulate matter, such as $PM_{2.5}$ and PM_{10} is of particular concern, as it represents the greatest health risk. These particles are formed through complex atmospheric reactions, often originating from domestic heating, industrial activities, and road transport (U.S. EPA, 2024). While air quality has improved across Europe in recent years, air pollution remains the leading environmental health hazard in the region (EEA, 2024). Cities in the Netherlands are no exception, where sectors like residential heating, road transport, and manufacturing contribute significantly to particulate matter pollution (EEA, 2023). Despite improvements, areas like The Hague continue to experience elevated pollution levels, exceeding acceptable thresholds for much of the year (IQAir, 2025). Monitoring air quality is critical to understanding and mitigating the effects of pollution. Accurate, reliable data is essential for informed decision-making, enabling policymakers to implement effective interventions. However, the quality of the data collected depends heavily on the reliability of the monitoring tools and sensors in use. Traditional air quality stations provide valuable data, but they are often limited in coverage, especially in densely populated or underserved areas. This highlights the importance of expanding monitoring networks with a diverse range of sensors, including both official stations and community-based sensors. Ensuring these sensors are strategically placed to cover all relevant areas is vital for accurate assessments. Gaps in coverage can lead to misleading conclusions and missed opportunities for intervention.

This report evaluates the air quality of one Dutch city, utilizing data from both official monitoring stations and community-based sensors. It assesses the reliability of these measurements and explores strategies for optimizing sensor placement to maximize coverage, with a particular focus on ensuring accurate air quality monitoring around schools.

1.1 Study Area

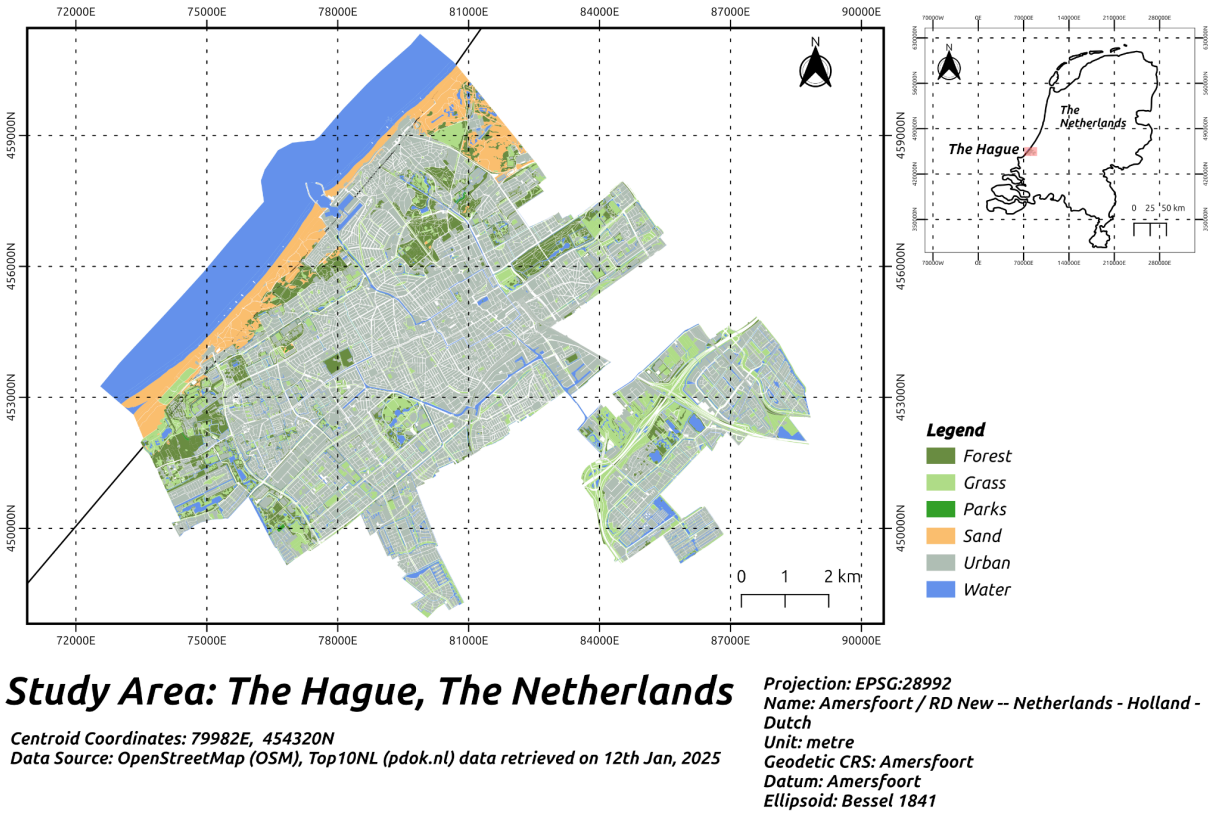


Figure 1 Study Area Map: The Hague Municipality and Land Use Patterns

This case study focuses on The Hague, a municipality and the capital of South Holland province in the Netherlands. It is the third-largest city in the country, with a population exceeding half a million(Gemeente Den Haag, 2024). While Amsterdam is the official capital of the Netherlands, The Hague serves as the administrative center and the seat of government(Gemeente Den Haag, 2024). The Hague is also one of the most densely populated municipalities in the Netherlands, with a population density of approximately 6,868 people per square kilometer, spread over an area of 9,820 hectares(Gemeente Den Haag, 2024). The majority of this area is built-up, leaving limited green cover. The western part of the city is characterized by coastal dunes and some green spaces. Traffic congestion is a notable issue in The Hague. According to the 2023 TomTom Traffic Report, residents spent an average of 1.5 hours more traveling within the city compared to the previous year(TomTom, 2023). This dense urban setup and high traffic volume significantly impact the environment, contributing to elevated pollution levels. Air pollution in The Hague is a pressing concern, with concentrations of pollutants frequently exceeding both European standards and World Health Organization (WHO) guidelines(EEA, 2023). This highlights the critical need for active monitoring and targeted interventions to identify and mitigate air quality hotspots.

1.2 Objectives

1. To evaluate the current reliability of the community sensor network in The Hague.
2. To develop an optimized placement strategy for air quality sensors to ensure maximum coverage of city schools.
3. To assess the improvement in reliability following the installation of new sensors.

2. Material & Methods

2.1 Methodology Workflow

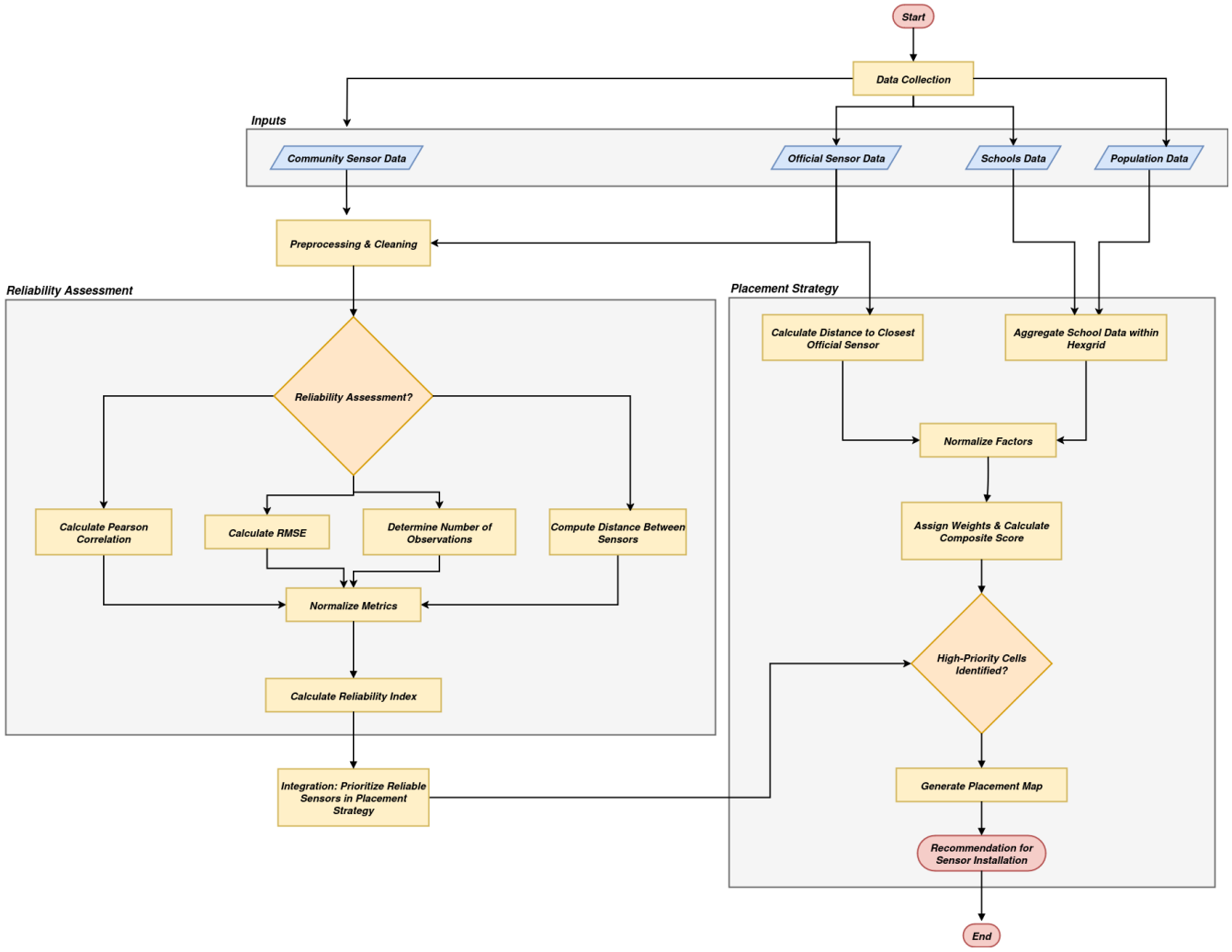


Figure 2 Methodology Flowchart: Steps for Reliability Assessment and Sensor Placement

2.2 Data Acquisition

To quantify air quality, this study focuses on two pollutants: $PM_{2.5}$ (particulate matter <2.5 µm) and PM_{10} (particulate matter <10 µm). Particulate matter consists of aerosol particles emitted primarily through anthropogenic activities or formed in the atmosphere via chemical reactions involving precursor gases like NOx and SOx(U.S. EPA, 2024). These pollutants were chosen for their severe and hazardous impact on human health. $PM_{2.5}$ is particularly dangerous as it penetrates deeply into the lungs. For perspective, an average human hair is approximately 70 µm in diameter—30 times larger than a $PM_{2.5}$ particle and 10 times larger than PM_{10}(U.S. EPA, 2024).

The World Health Organization (WHO) has established guidelines for daily and annual limits for $PM_{2.5}$ and PM_{10} emissions(EEA, 2022). The European Union (EU) also sets its own standards, which are more flexible but broadly align with WHO recommendations(EC).

Pollutant	**WHO Limits (µg/m³)**	**EU Limits (µg/m³)**
$PM_{2.5}$ (Annual)	5	25
$PM_{2.5}$ (24-Hour)	15	Not Specified
PM_{10} (Annual)	15	40
PM_{10} (24-Hour)	45	50 (max 35 exceedances/year)

Table 1 Description of WHO and EU limits on PM_{10} and $PM_{2.5}$ concentrations daily and annually.

2.2.1 Official Air Quality Monitoring

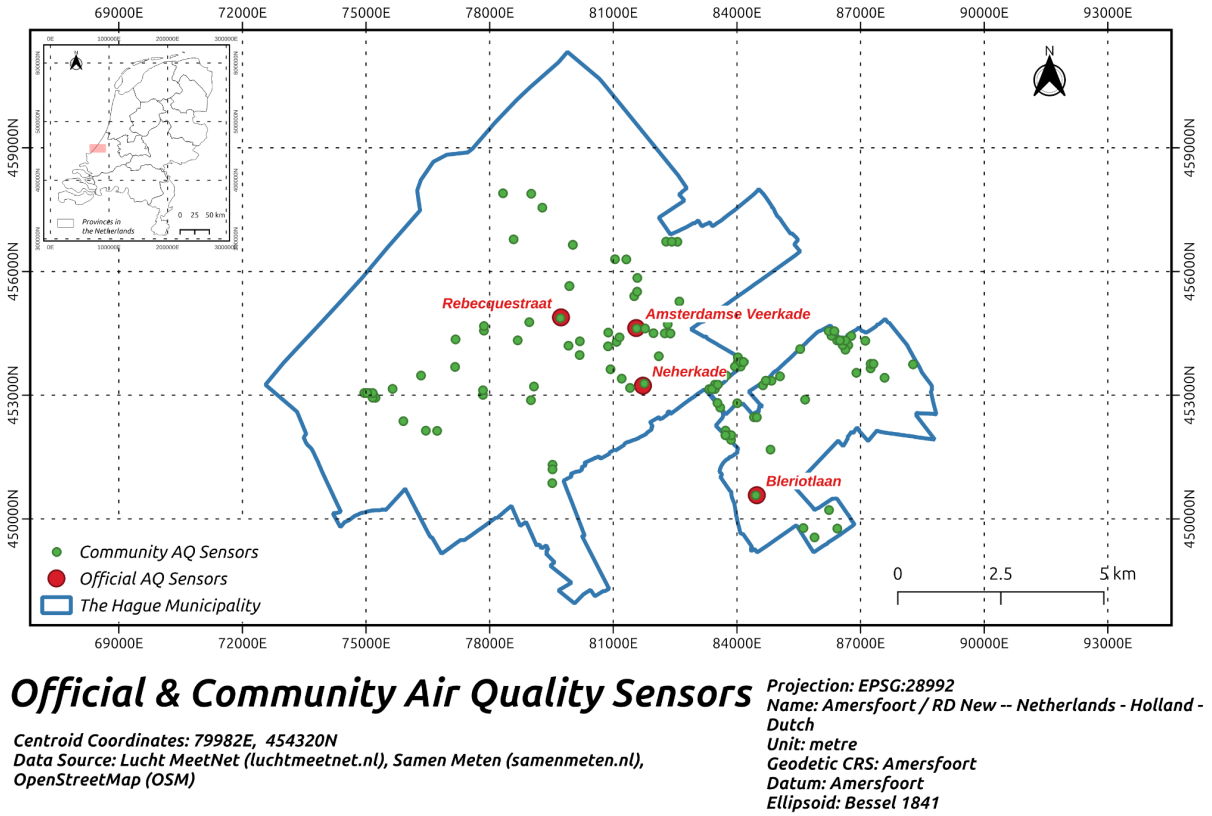
In the Netherlands, air quality is monitored through a collaboration of national agencies, including the National Institute of Public Health and the Environment (RIVM), GGD Amsterdam, the Environmental Protection Agency (EPA), and DCMR Rotterdam(IQAir, 2025). These organizations utilize a combination of high-precision reference-grade instruments and low-cost sensors. Data is accessible via the air quality portal Luchtmeetnet http://www.luchtmeetnet.nl/ in real-time and as archival datasets. As of January 2025, there are approximately 101 official monitoring stations across the Netherlands, of which four are located in The Hague. All four measure PM_{10}, while only two measure $PM_{2.5}$. Sensor metadata and coordinates were retrieved using the Luchtmeet API https://api-docs.luchtmeetnet.nl/, while measurement data for the entire year was sourced from a different archival data portal https://data.rivm.nl/data/luchtmeetnet/ due to the high API request limit and slow response times. A total of six years of hourly data (2019–2024) was collected to ensure sufficient temporal coverage, even accounting for gaps.

Sensors	**Sensor ID**	**Pollutants**	**Type**	**Organisation**
Rebecquestraat	NL10404	PM_{10}, $PM_{2.5}$	Municipal	RIVM
Amsterdamse Veerkade	NL10445	PM_{10}	Traffic	RIVM
Bleriot Laan	NL10446	PM_{10}	Municipal	RIVM
Neherkade	NL10450	PM_{10}, $PM_{2.5}$	Traffic	RIVM

Table 2 Details of official sensors, including their type, managing organization, and the pollutants they measure.

2.2.2 Community Air Quality Monitoring

The Netherlands also has an extensive network of low-cost sensors maintained by the community, supported and calibrated by RIVM under the "Samen Meten" initiative https://www.samenmeten.nl/. As of January 2025, approximately 19,584 community sensors operate across the country, with 140 located within the study area. By applying a proximity threshold of 500 m (and later extending to 2 km to address data gaps), 6 sensors were identified and stored in a shapefile with CRS EPSG:28992. For each sensor, six years of measurement data (2019–2024) was downloaded and saved as CSV files for further analysis.



Official & Community Air Quality Sensors

Centroid Coordinates: 79982E, 454320N
Data Source: Lucht MeetNet (luchtmeetnet.nl), Samen Meten (samenmeten.nl), OpenStreetMap (OSM)

Projection: EPSG:28992
Name: Amersfoort / RD New -- Netherlands - Holland - Dutch
Unit: metre
Geodetic CRS: Amersfoort
Datum: Amersfoort
Ellipsoid: Bessel 1841

Figure 3 Current Official and Community Sensors Within The Hague Municipality

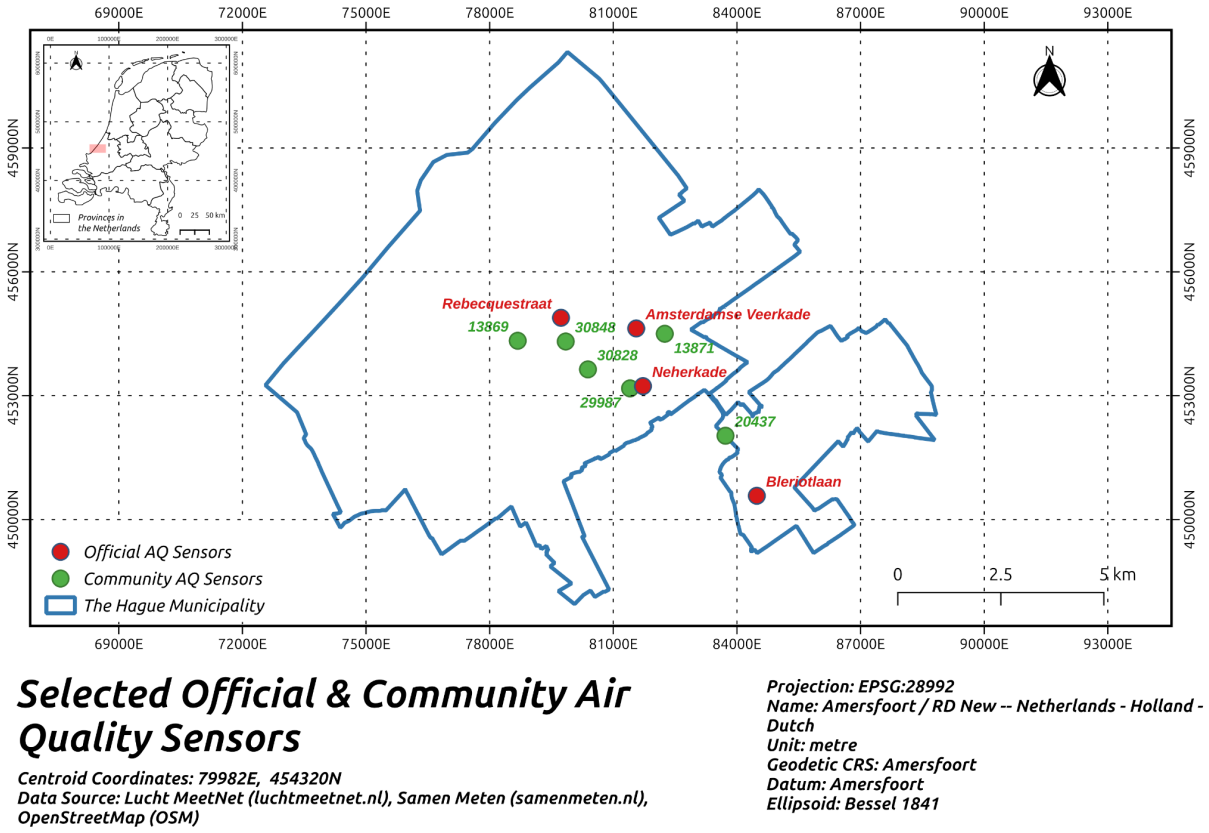
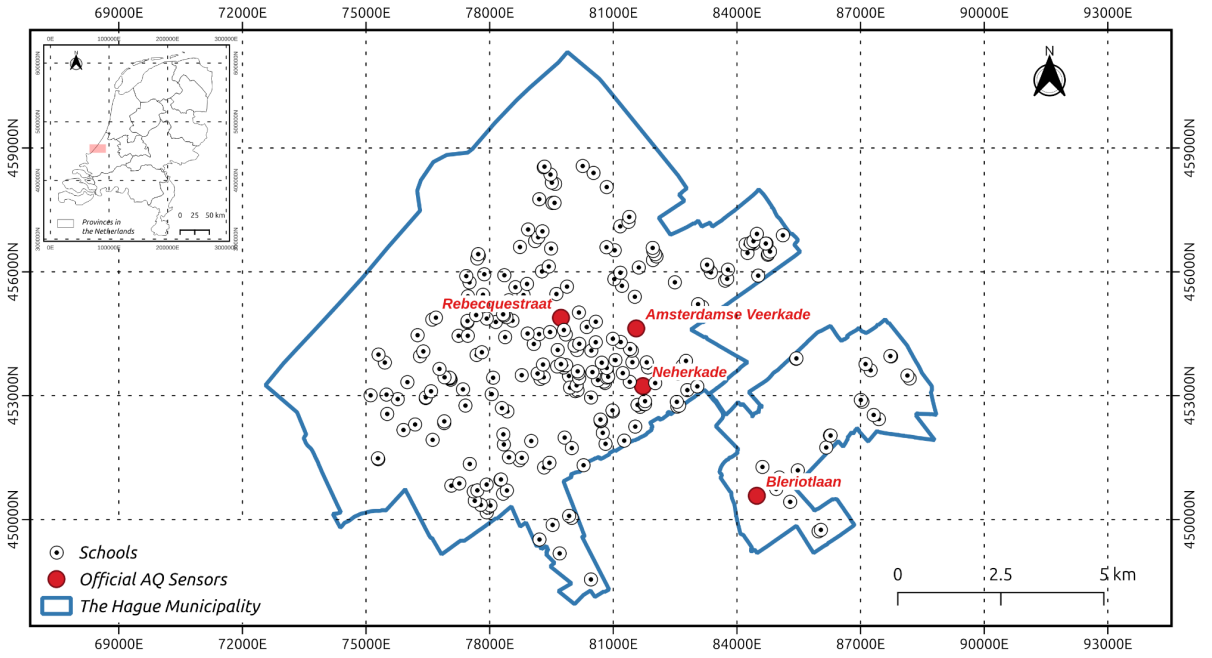


Figure 4 Selected Official and Community Air Quality Sensors in The Hague

2.2.3 Schools

School locations within the study area were extracted from OpenStreetMap (OSM) https://overpass-turbo.eu/ using the tag "amenity=school." A total of 251 schools were identified and stored as a shapefile with CRS EPSG:28992.

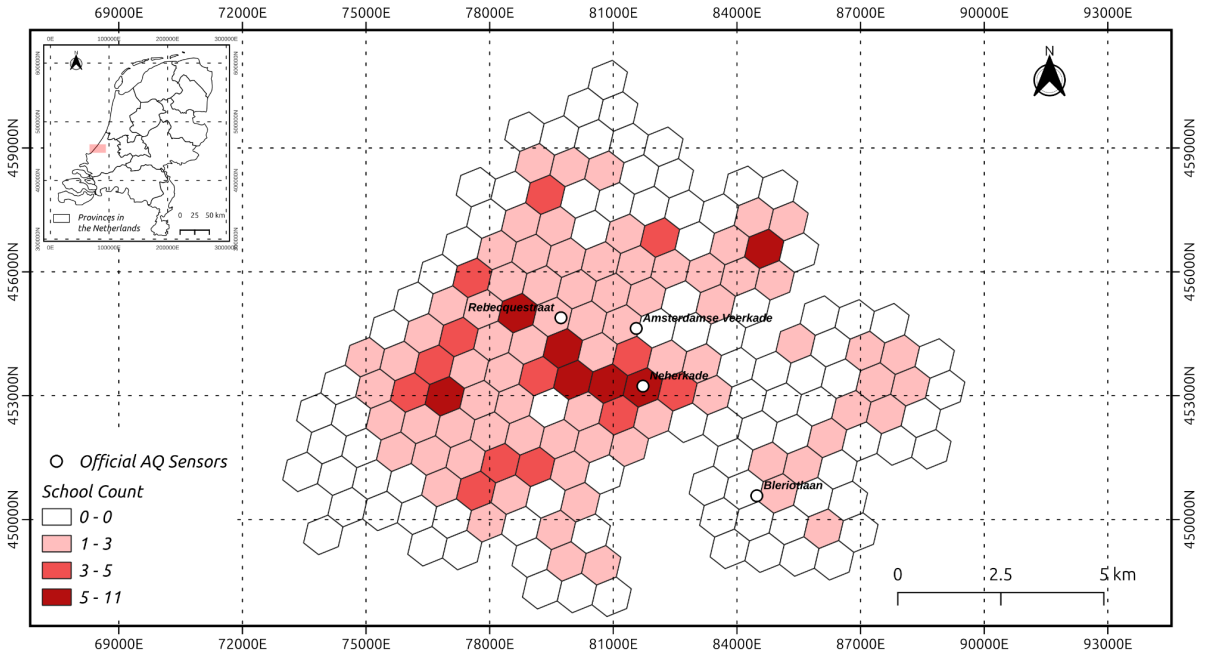


Official Air Quality Sensors & Schools

Centroid Coordinates: 79982E, 454320N
Data Source: Lucht MeetNet (luchtmeetnet.nl), OpenStreetMap (OSM)

Projection: EPSG:28992
Name: Amersfoort / RD New -- Netherlands - Holland - Dutch
Unit: metre
Geodetic CRS: Amersfoort
Datum: Amersfoort
Ellipsoid: Bessel 1841

Figure 5 School Locations and Official Sensor Sites Within The Hague Municipality



Distribution of Schools in 500m Hexagonal Grid Cells

Centroid Coordinates: 79982E, 454320N
Hex Grid: ~500m (Side length)
Data Source: Kontur Population Dataset (kontur.io/portfolio/population-dataset)

Projection: EPSG:28992
Name: Amersfoort / RD New -- Netherlands - Holland - Dutch
Unit: metre
Geodetic CRS: Amersfoort
Datum: Amersfoort
Ellipsoid: Bessel 1841

Figure 6 Distribution of Schools in 500m Hexagonal Grid Cells Alongside Official Sensor Locations in The Hague

2.2.4 Population

Population data was acquired from Kontur Population. It is represented by H3 hexagons with population counts at 500m resolution. The dataset is primarily developed using Global Human Settlement Layer (GHSL) but has also integrated Facebook population data (HRSL), Microsoft Building Footprint, Land Information New Zealand, and Copernicus Global Land Service data to improve distribution accuracy(Kontur, 2023).

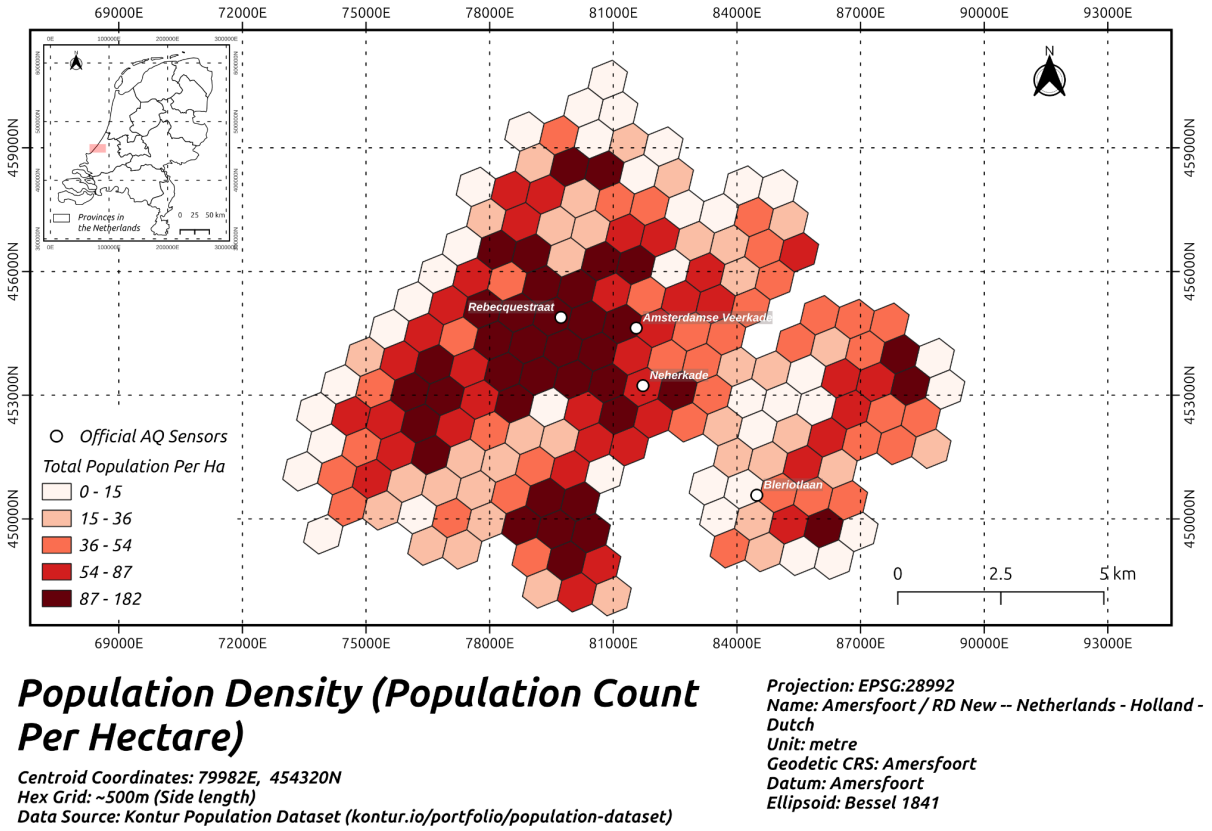


Figure 7 Population Density Map of The Hague Overlaid with Official Sensor Locations

2.3 Methodology

Hourly measurements from both official and community sensors were aggregated to daily frequency. Community sensor data was merged with the nearest official sensor data for corresponding dates. This synchronization of daily measurements enabled direct comparison. However, due to the non-standard temporal coverage of community sensors, a significant portion of official sensor data had to be dropped where matching data was unavailable.

2.3.1 Defining "Reliability"

Reliability was evaluated through a comprehensive analysis of the agreement between official and community sensor measurements. Four key factors were considered in this evaluation: the Pearson correlation coefficient (r), the number of observations, root mean square error (RMSE), and the distance between sensors. Each factor was selected to address specific aspects of measurement quality and consistency. The Pearson correlation coefficient (r) was computed to quantify the strength of the linear relationship between the official and community sensor measurements. A higher r value indicates stronger agreement, reflecting the ability of community sensors to capture trends similar to official sensors. Root mean square error (RMSE) was used to measure the magnitude of deviation between the two datasets. Lower RMSE values indicate closer agreement between the measurements, highlighting sensors with higher accuracy. The number of observations was included as it directly affects

the statistical robustness of the analysis. A higher number of observations reduces the influence of random errors or outliers, providing greater confidence in the reliability assessment. Finally, the distance between sensors was considered to account for spatial effects. Sensors located closer together are more likely to experience similar environmental conditions, leading to better agreement, while larger distances can introduce variability due to localized factors. To enhance the evaluation, a composite reliability index was developed by integrating these factors. The weights were assigned based on their relative importance: 0.3 for pearson correlation, 0.3 for RMSE, 0.2 for number of observations and 0.2 for distance between sensors. This index provides a robust measure of reliability by balancing the strength of agreement, data availability, accuracy, and spatial proximity, ensuring that sensors with both high agreement and sufficient data are prioritized for further analysis and calibration.

Reliability was defined as follows:

1. Pearson Correlation (r): Pearson Correlation was normalised between [0,1] using the below formula

$$r_{norm} = \frac{r+1}{2}$$

2. Number of Observations (n): The number of observations were normalised using min-max normalisation.

$$n_{norm} = \frac{n-n_{min}}{n_{max}-n_{min}}$$

3. Distance between sensors (d): The distance between sensors were normalised inversely to prioritise closer proximity.

$$d_{norm} = 1 - \frac{d-d_{min}}{d_{max}-d_{min}}$$

4. Root Mean Square Error (RMSE): Root mean square error was computed between official & community measurements. Higher RMSE signifies lower reliability.

$$rmse_{norm} = 1 - \frac{MSE-MSE_{min}}{MSE_{max}-MSE_{min}}$$

5. Weighting Factors: A weighted sum of normalise factors was computed, with weights assigned to reflect their relative importance (for eg., for correlation, for observation, for distance, for RMSE)

$$Reliability\ Index = w_1 * r_{norm} + w_2 * n_{norm} + w_3 * d_{norm} + w_4 * rmse_{norm}$$

2.3.2 Defining "Placement Strategy"

To identify optimal locations for new air quality sensors, a weighted scoring approach similar to the reliability index was employed, incorporating population, number of schools, and the existing official sensor network. Schools were prioritized as they are critical environments for children, who are more vulnerable to air pollution, and represent areas with consistent human activity. Population density was included to ensure sensors serve the maximum number of

residents, capturing the most significant exposure risks. Distance to existing sensors was considered to avoid redundancy and ensure broader spatial coverage, improving the overall network efficiency. Population data, represented in hexagons grid with a 500m resolution, served as the baseline for population counts. School data was overlaid, and a point-in-polygon query was conducted to aggregate the number of schools within each grid cell. Additionally, the distance to the nearest official sensor was calculated from the centroid of each grid cell. The resulting dataset contained values for total population, school count, and distance to the nearest official sensor for each grid cell. Each factor was normalized using min-max normalization to scale values between 0 and 1. A composite placement score was then calculated by taking the weighted sum of these factors. The weights were assigned based on their relative importance: 0.5 for school count, 0.3 for population count, and 0.2 for distance to the closest sensor. The hex grid was visualized based on the composite scores, with the four existing official sensors overlaid to evaluate current school coverage. Grid cells with high composite scores (> 0.48, chosen based on the quantile distribution) were prioritized as candidate locations for six new official sensors. Cells with lower composite scores (<= 0.48 & > 0.3) were identified as suitable for community sensors to ensure broader spatial coverage.

Placement strategy was defined as follows:

1. School coverage: School count was normalized in the range of 0-1. Areas with a higher number of schools were prioritized.

$$school_{norm} = \frac{(school - school_{norm})}{(school_{max} - school_{min}}$$

2. Population Coverage: Areas with large populations were prioritized to monitor exposure.

$$population_{norm} = \frac{population - population_{min}}{population_{max} - population_{min}}$$

3. Distance to existing official sensor: Distance to the nearest official sensor from grid cell was calculated.

$$distance_{norm} = \frac{distance - distance_{min}}{distance_{max} - distance_{min}}$$

4. Weighting Factor (w_s, w_p, w_d): Weights were assigned to each factor based on its importance such that $w_s + w_p + w_d = 1$, where ws, wp, wd denotes the weights assigned to schools, population and distance.

$$Composite\ Placement\ Score = w_s * school_{norm} + w_p * population_{norm} + w_d * distance$$

2.3.3 Reliability Scores: Pre- and Post-Sensor Installation

To calculate the reliability scores for the entire sensor network, the average reliability scores of six community sensors and four official sensors (selected for the study) were used. Official

sensors were assumed to have a reliability score of 0.99, while the scores for community sensors were calculated using the previously described reliability methodology. After placing the new official and community sensors using the placement strategy, reliability scores were recalculated. The reliability scores for the official sensors remained constant at 0.99, while the scores for the new community sensors were calculated as the sum of the average reliability score of the existing community sensors and the product of a distance weight (0.20) and the normalized distance between the new community sensor and its nearest official sensor. Finally, the overall network reliability score was recalculated by taking the average of the reliability scores for both the official and community sensors after the addition of the new sensors.

$$ReliabilityScore_{Network}(Before) = \frac{(Sum(ReliabilityScore_{official}(Before)) + Sum(ReliabilityScore_{community}(Before))}{Num_{Official}(Before) + Num_{Community}(Before)}$$

$$ReliabilityScore_{community}(After) = \frac{Sum(ReliabilityScore)_{community}(Before)}{Num_{community}(Before)} + w_d * (1 - \frac{d - d_{min}}{d_{max} - d_{min}})$$

$$ReliabilityScore_{Network}(After) = \frac{(Sum(ReliabilityScore_{official}(After)) + Sum(ReliabilityScore_{community}(After))}{Num_{official}(After) + Num_{community}(After)}$$

where $ReliabilityScore_{Network}(Before)$, $ReliabilityScore_{Network}(After)$ is the reliability scores of the entire network before and after installing the new sensors. $Num_{official}$ and $Num_{community}$ is the number of official and community sensors.

3. Results

3.1 Reliability Assessment

Pollutant	Official Sensor	Community Sensor	Number of Observations	Distance From Official Sensor(Km)	r(PCC)	Reliability Index	p<0.001	RMSE(ug/m3)
$PM_{2.5}$	NL10404	30848	699	0.586	0.90	0.73	1.29e-249	4.08
$PM_{2.5}$	NL10450	30828	255	1.240	0.87	0.58	2.43e-81	5.03
PM_{10}	NL10404	13869	457	1.188	0.84	0.63	5.70e-123	5.45
PM_{10}	NL10445	13871	364	0.704	0.91	0.62	6.15e-139	5.07
PM_{10}	NL10446	20437	104	1.641	0.86	0.52	3.62e-32	6.25
PM_{10}	NL10450	29987	41	0.317	0.91	0.53	4.58e-16	4.85

Table 3 Summary of Reliability Metrics for $PM_{2.5}$ and PM_{10} Measurements: Analysis of Official and Community Sensor Pairings, Including Correlation (r), RMSE, and Reliability Index Based on Spatial Proximity and Data Availability

To evaluate the performance and reliability of community air quality sensors relative to official sensors, key metrics were analyzed, including mean, RMSE, Pearson correlation coefficient, reliability index, and the number of observations. Table 3 provides a summary of the comparison of $PM_{2.5}$ and PM_{10} concentrations measured by community and official sensors, highlighting variations in performance and data quality. The Pearson correlation (r) values range from 0.84 to 0.91 for PM_{10} and 0.87 to 0.90 for $PM_{2.5}$, indicating a strong agreement between community and official sensors. The reliability index decreases with increasing distance from the official sensor. For instance, the $PM_{2.5}$ sensor at 0.586 km has a reliability index of 0.73, which decreases to 0.58 at 1.240 km. A similar trend is observed for PM_{10}, where sensors farther away have lower reliability scores. The number of observations also appears to influence reliability. Sensors with a larger number of observations, such as 699, 457 and 364 tend to have higher reliability scores 0.73, 0.63 and 0.62 respectively while those with fewer observations like 41 and 104, exhibit lower reliability. RMSE values increase with decreasing reliability, indicating poorer accuracy in the measurement of community sensors compared to official sensors. For example, the sensor with the lowest reliability index (0.52) for PM_{10}, has an RMSE of 6.25 ug/m3, the highest in the table. Although still preliminary and limited in scope, this analysis highlights how distance, observation count, and RMSE influence the reliability index, helping identify sensors that may require calibration or further attention. Furthermore, time-series trends revealed a notable pattern: for most sensor pairs, community sensors tended to overestimate concentrations compared to official ones (±5ug/m3) as shown in figure 9-12. This discrepancy likely arises from the higher sensitivity of low-cost sensors to

their immediate surroundings, emphasizing the need for additional calibration to improve accuracy.

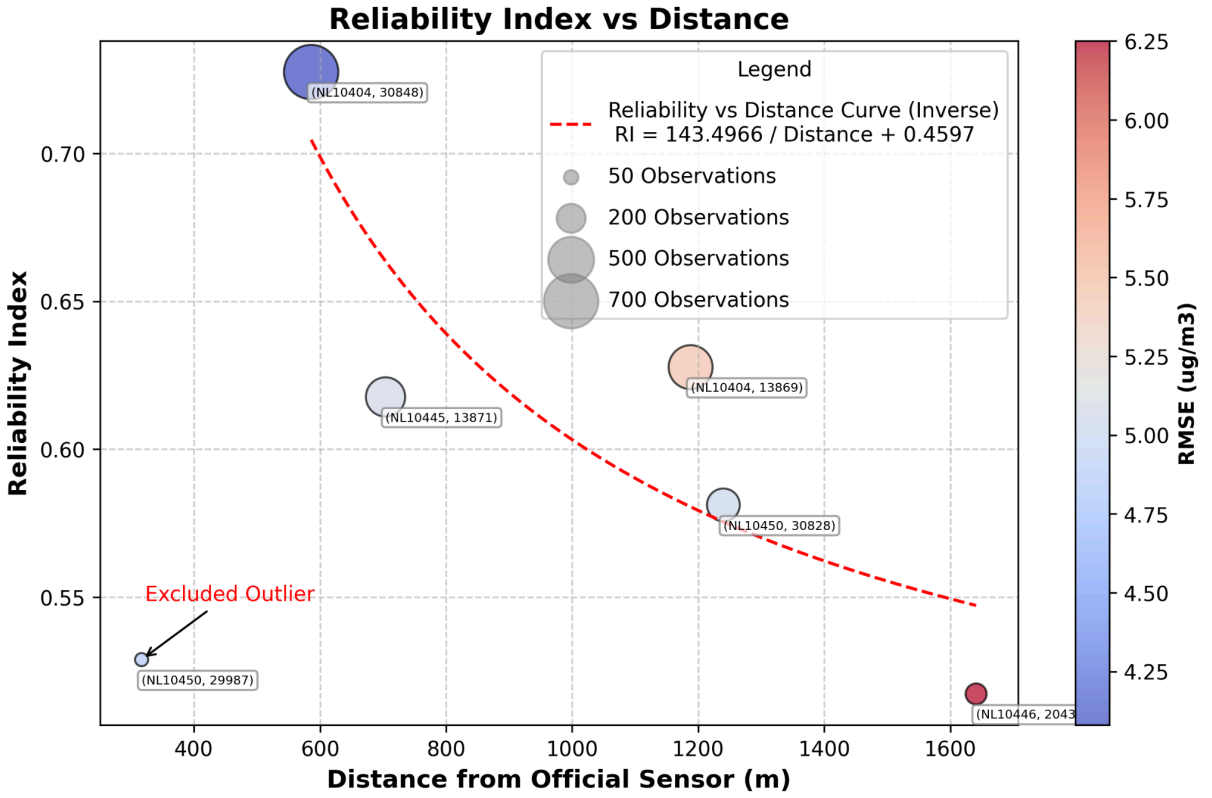


Figure 8 Relationship Between Reliability Index and Distance from Official Sensors: Influence of Observation Count and RMSE on Data Reliability

Figure 8 shows the analysis between reliability index and distance to closest to official sensor. The analysis reveals an inverse relationship between the Reliability Index (RI) and the Distance from Official Sensors, as demonstrated by the fitted curve $RI = \frac{Distance}{143.4966} + 0.4597$. This relationship indicates that reliability decreases with increasing distance from the official air quality sensors. Observations with higher reliability are typically located closer to the sensors, while measurements at greater distances exhibit reduced reliability and higher RMSE values. The results align with Tobler's First Law of Geography, which states that "everything is related to everything else, but near things are more related than distant things"(Tobler,1970). However, the relationship is not highly significant, as some measurements at closer distances still exhibit lower reliability, likely due to local factors or outlier effects. One notable outlier at approximately 400 meters, with unusually low reliability(0.53) despite proximity to a sensor, was excluded from the curve fitting due to having too few observations (41) compared to other sensors (>100). Since the number of observations is a critical factor in ensuring the reliability of the analysis, the small sample size of the outlier group led to its exclusion, as it lacked statistical stability compared to the larger groups. These findings highlight the need to consider spatial relationships and local environmental influences when evaluating sensor placement and measurement accuracy.

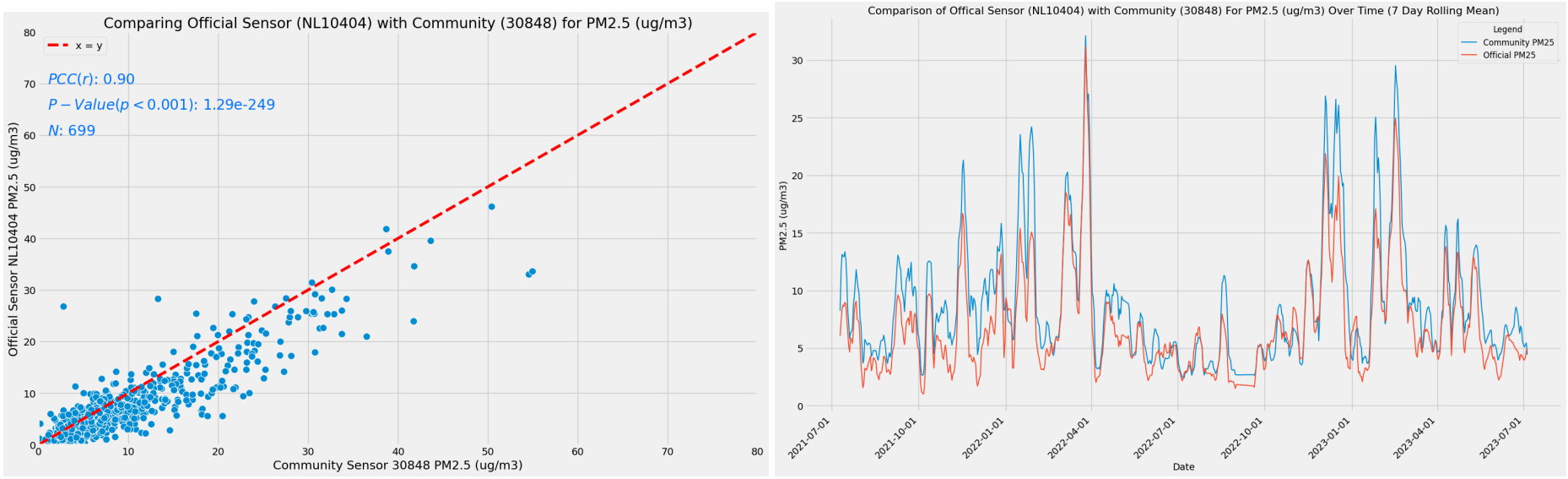


Figure 9 Comparison of Official (NL10404 Rebecquestraat) and Community Sensor (30848): Scatter Plot and Time Series Analysis

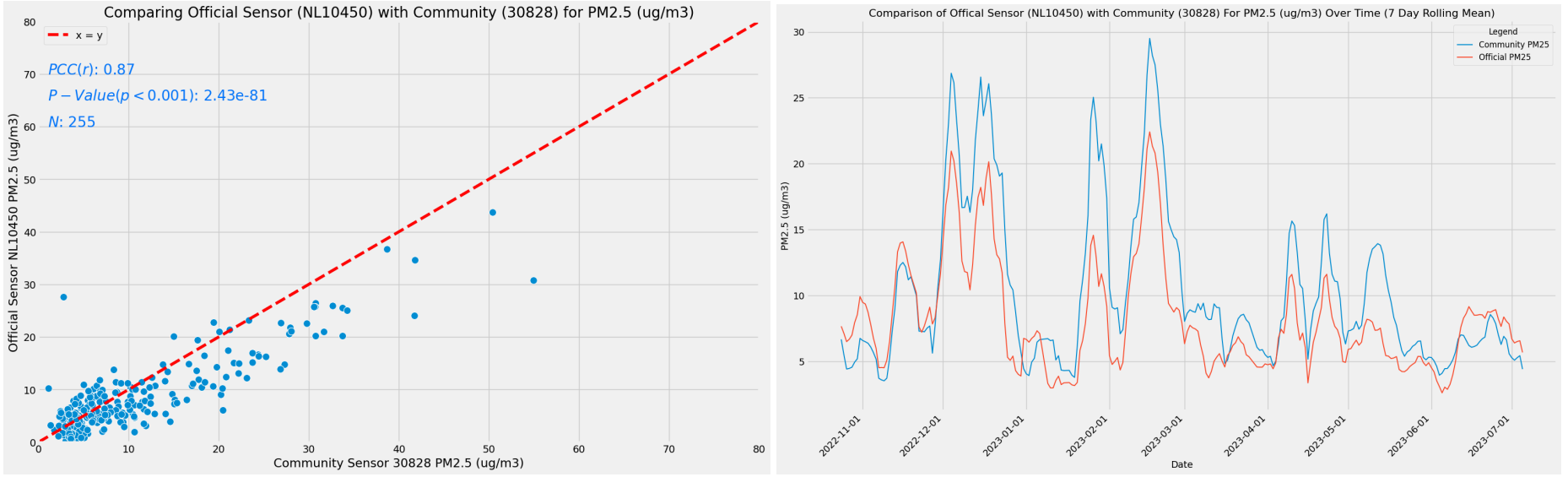
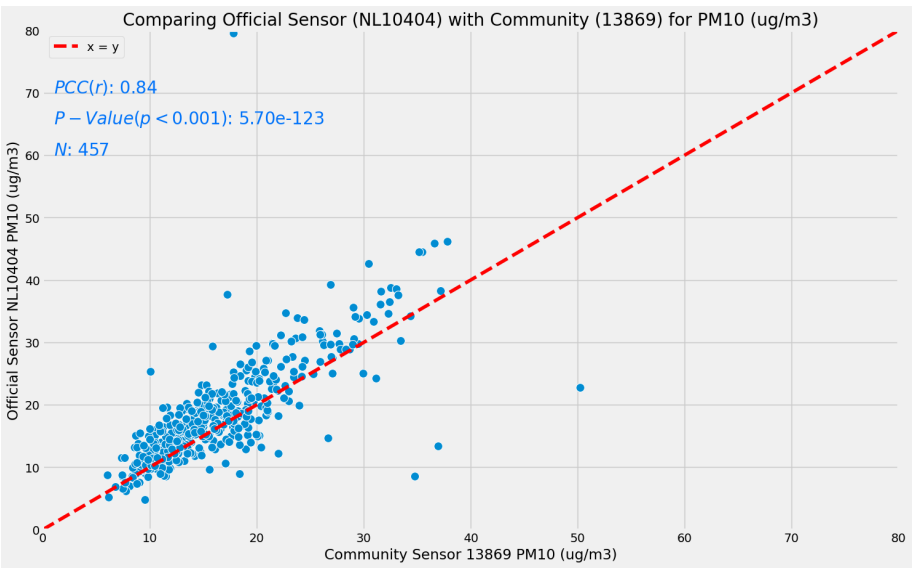


Figure 10 Comparison of Official (NL10450 Rebecquestraat) and Community Sensor (30828): Scatter Plot and Time Series Analysis



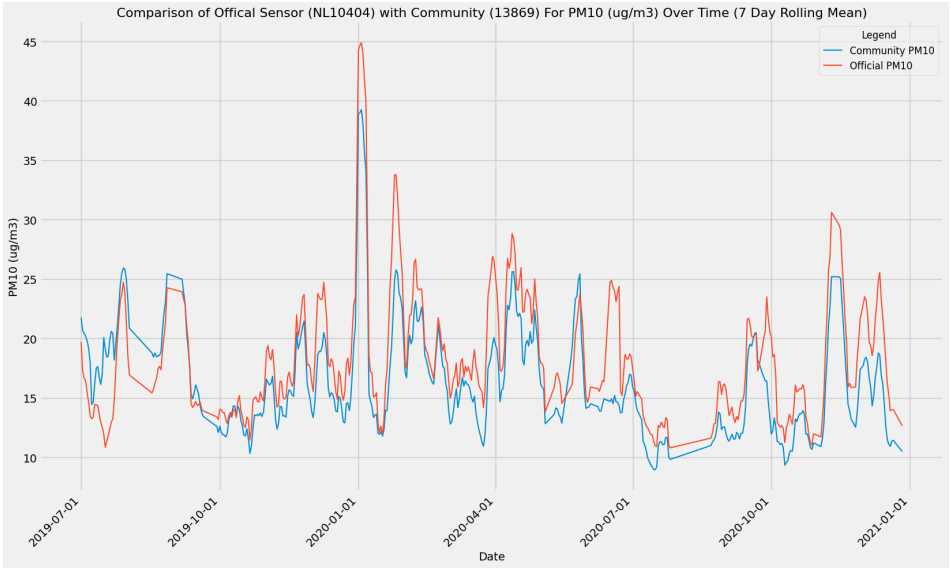
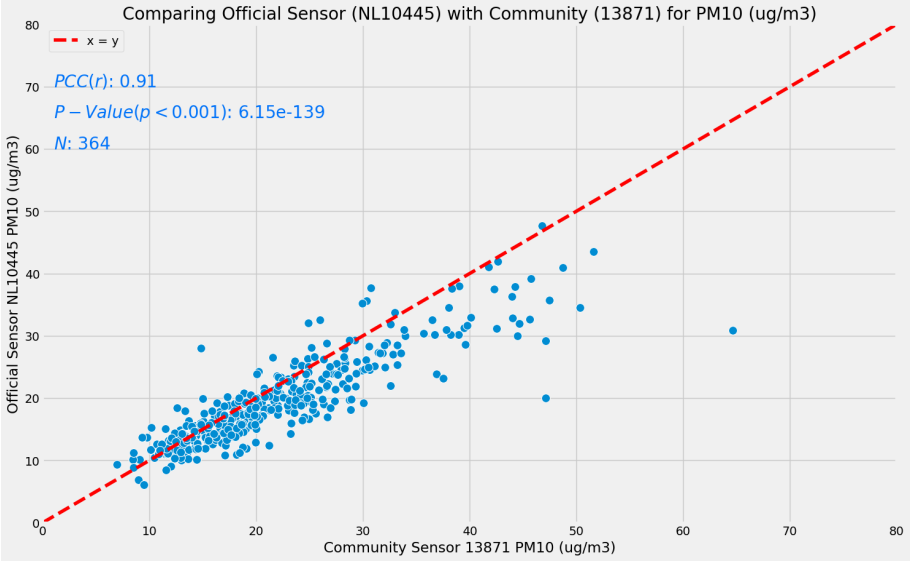


Figure 11 Comparison of Official (NL10404 Rebecquestraat) and Community Sensor (13869): Scatter Plot and Time Series Analysis



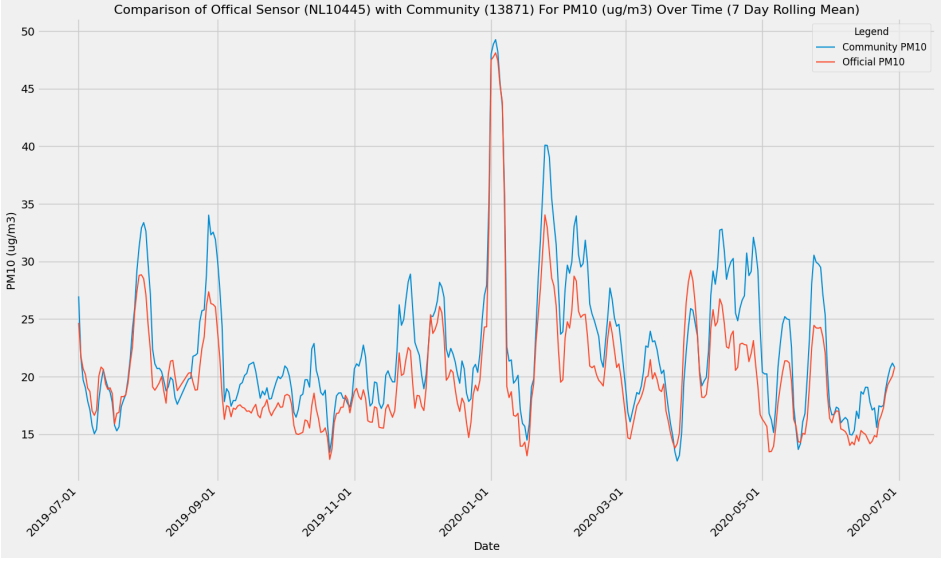


Figure 12 Comparison of Official (NL10445 Rebecquestraat) and Community Sensor (13871): Scatter Plot and Time Series Analysis

3.2 Placement Strategy

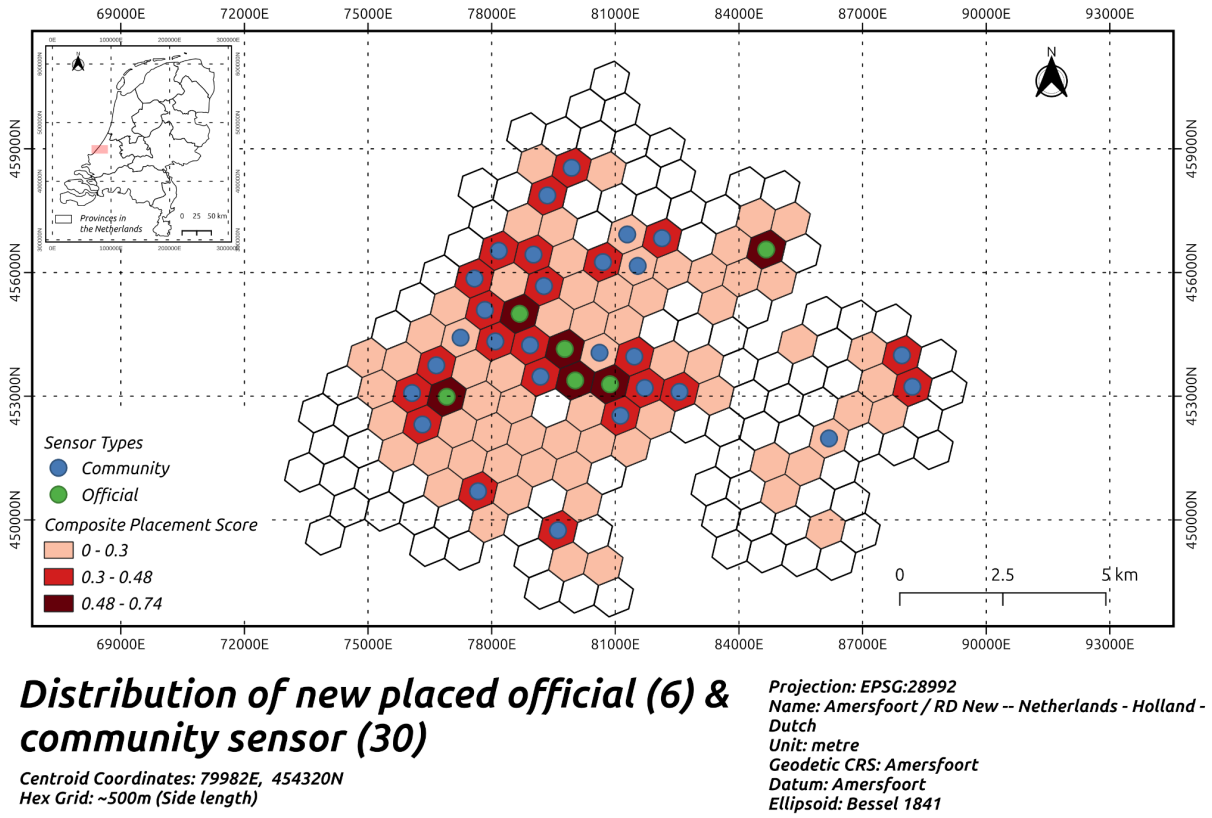


Figure 13 Final Placement of Official and Community Air Quality Sensors in The Hague with Grid Cell Distribution

To address spatial gaps in air quality monitoring, a placement strategy was devised using a composite score based on population density, school density, and proximity to existing sensors. Figure 13 visualizes optimal locations for six new official air quality sensors and 30 community sensors. High-priority grid cells with composite scores ≥ 0.48 were selected for the 6 official sensors, while cells with lower scores between 0.3 and 0.48 were deemed suitable for additional 30 community sensors. The placement map highlights high-priority areas such as the Centrum region, where high population and school densities coincide with a lack of existing sensor coverage. This strategy ensures enhanced monitoring in densely populated and vulnerable areas, particularly schools, which house children most at risk from air pollution.

3.3 Reliability Scores: Pre- and Post-Sensor Installation

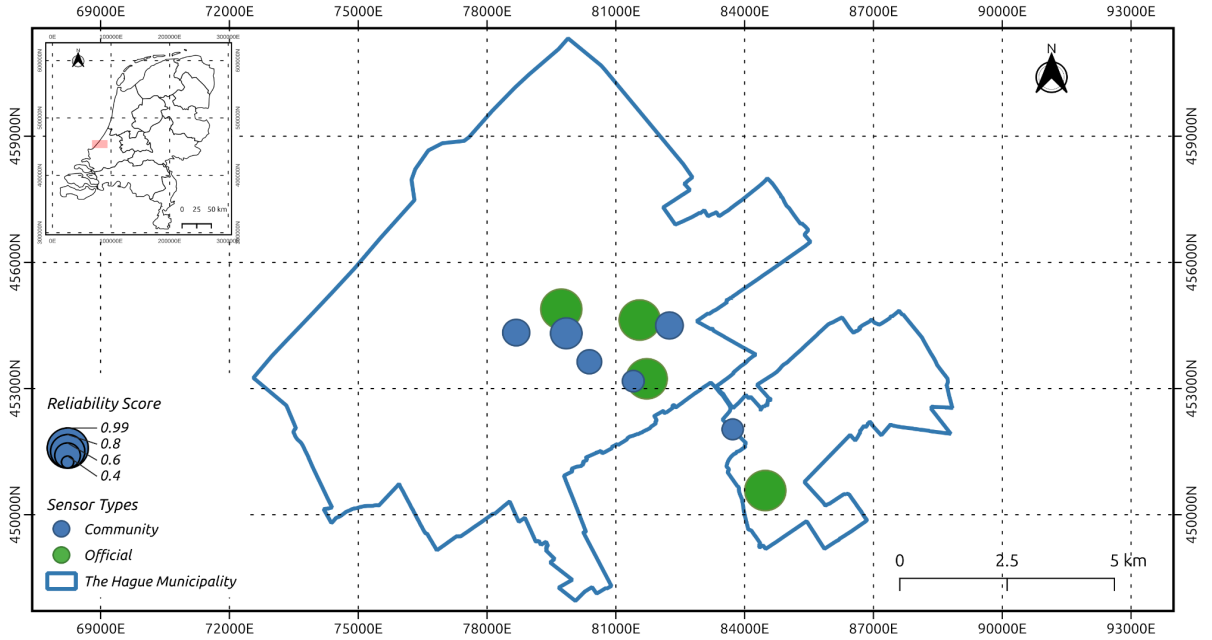


Figure 14 Updated Reliability Scores Before Sensor Network Expansion

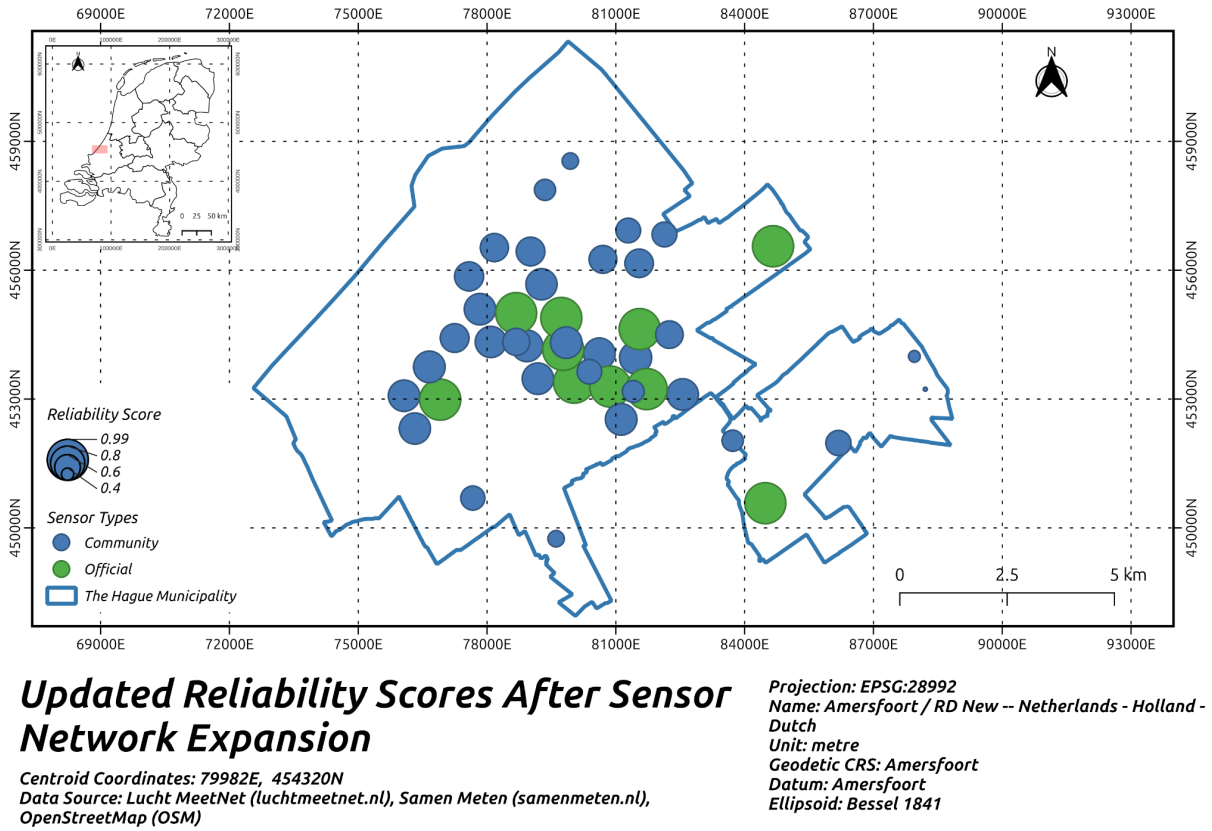


Figure 15 Updated Reliability Scores After Sensor Network Expansion

The initial reliability score of the entire sensor network, which comprised four official sensors and six community sensors as shown in Figure 14, was calculated as 0.76. The average reliability score of the official sensors was 0.99, while the community sensors had an average reliability score of 0.613. After the installation of additional sensors, the network expanded to include 10 official sensors (six newly installed) and 36 community sensors (five new sensors placed around each new official sensor in addition to the six existing ones) as shown in Figure 15. The updated reliability score for the expanded network was calculated by the average of the official sensor scores (0.99) and community sensors, resulting in a final score of 0.72, showing a slight decrease compared to the initial score. This drop in post-expansion reliability can be attributed to the addition of more community sensors, which generally have lower reliability compared to official sensors. Furthermore, the placement of community sensors prioritized covering larger populations and more schools, even if it meant placing some sensors farther from official sensors. Before the addition of new sensors, the network covered 15% of schools and 8% of the population. After the expansion, coverage significantly increased to 60% of schools and 36% of the population. While this improved the network's spatial coverage and potential impact, the expanded placement also contributed to a slight reduction in overall reliability, as some community sensors were located farther from official sensors.

4. Discussion

This section discusses the results of the case study and their interpretations.

4.1 Reliability Assessment

Reliability is a critical consideration when working with air quality sensors, especially low-cost sensors, as their measurements can be influenced by environmental factors, sensor drift, and calibration inaccuracies. For low-cost sensors, reliability assessments often involve comparing their outputs with reference-grade instruments to gauge the accuracy and consistency of their readings under varying conditions. The reliability indicator was computed using Pearson's correlation, RMSE, the number of observations, and the distance between sensors. The results exhibited variation across different sensor pairs, reflecting the influence of the selected input indicators. The analysis identified patterns where sensors with higher correlation, lower error metrics, closer proximity to official sensors, and fewer missing data points were considered more reliable. For example, in the case of $PM_{2.5}$ measurements, one community sensor achieved a reliability index of 0.73, while another scored 0.58. In contrast, for PM_{10}, the reliability index ranged from 0.63 to 0.52, which was lower than for $PM_{2.5}$. These findings suggest that the chosen indicators effectively captured both spatial (distance) and temporal (long-term data series) factors, as well as the accuracy of the measurements (RMSE and correlation), reinforcing the trustworthiness and relevance of the data quality. The proposed reliability index can be adapted to include additional environmental indicators that affect air quality, enhancing its robustness.

4.2 Placement Strategy

Optimal placement of air quality sensors is critical to improving spatial coverage and eliminating gaps in air quality monitoring. It also prevents redundancy and unintentional bias introduced by placing sensors too close to each other. This study presents a framework that utilizes a weighted sum of indicators—school density, population density, and proximity to existing sensors—to classify optimal locations for sensor placement. These indicators highlight the densest areas of society, where the majority of the population resides and schools, which house children most vulnerable to air pollution, are located. Based on the weighting mechanism, the algorithm identified areas with poor or no coverage from existing official air quality sensors. For example, areas near Centrum, which have the highest population and school density, are currently not covered by any official sensors. Similar patterns were observed across the municipality. The study proposes installing six additional official sensors in cells with the highest composite scores (>0.48) and deploying 30 additional community sensors in surrounding cells with scores between 0.3 and 0.48. This framework provides a robust foundation for improving air quality monitoring while covering the majority of the population and schools. It can be further optimized and adapted by integrating other indicators, such as road networks and industrial versus residential areas.

4.3 Reliability Scores: Pre- and Post-Sensor Installation

The expansion of the sensor network to cover larger populations and more schools resulted in a slight decrease in overall reliability, dropping from 0.76 to 0.72. Before the addition of new sensors, the network covered only 15% of schools and 8% of the population. Following the expansion, coverage significantly improved, reaching 60% of schools and 36% of the population. However, this improvement in spatial coverage came at the cost of a slight

reduction in overall reliability due to the addition of more community sensors and their placement farther from official sensors. To improve future reliability scores, the network design could incorporate strategies such as optimizing the placement of community sensors closer to official sensors. Additionally, introducing more official sensors in high-priority areas could further strengthen the network's overall reliability.

4.4 Choice of cartographic techniques

The study used various cartographic techniques to highlight key findings. Sensor locations, both official and community, were shown as spatial points with hue, following Bertin's visual variables for categorical data, allowing clear differentiation. The distribution of schools and population was represented in hexagonal grids ensuring uniform representation with color indicating school bins and population density. This method simplifies large datasets and reveals spatial trends without visual clutter. The reliability vs. distance map used scatter plots with marker size representing the number of observations and color based on RMSE, making it easy to assess data reliability. Sensor placement was shown with hue for sensor type, overlaid with a choropleth layer for placement scores, ensuring a clear distinction between official and community sensors while also showing placement strategy. Lastly, reliability scores of additional sensors were visualized with proportional markers, with size reflecting reliability and hue distinguishing sensor types. In conclusion, the chosen cartographic techniques effectively communicate complex spatial relationships and data, making the study's findings accessible and understandable.

4.5 Limitations

This study is not without its constraints. Temporal gaps and inconsistencies in the observations of community sensor data posed significant challenges during analysis. Many community sensors either lacked data for extended periods or provided identical data without clarity about the source. The weighting scheme for the reliability index is based on predefined indicators with constant weights, which may introduce bias. Dynamic weight assignment based on spatial and temporal contexts could improve the framework. Similarly, normalization parameters might vary across regions and cities, which could limit generalizability. Finally, the spatial generalizability of the current strategy to cities with different population distributions and air quality characteristics remains untested.

5. Conclusion

This study demonstrates a comprehensive approach to improving air quality monitoring by integrating a reliability index and an optimized placement strategy. By utilizing key indicators, such as correlation, RMSE, the number of observations, and proximity, the reliability index effectively highlights the quality of sensor data. The placement strategy ensures efficient sensor deployment, addressing gaps in spatial coverage and prioritizing vulnerable populations.

Future work aims to scale and expand this proof of concept by:

1. Refining the reliability index with additional factors, such as weather data and terrain information.
2. Validating the placement strategy in new cities and regions.
3. Developing dynamic models for weight assignment.
4. Including additional factors, such as traffic patterns, industrial emissions, and time-of-day effects.
5. Extending the methodology to cover other key locations, such as hospitals and playgrounds.

By addressing these aspects, the framework can become a more adaptable and comprehensive solution for air quality monitoring, benefiting public health and urban planning efforts.

6. References

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End of report. Appendix on next page

Appendix

Data Management Plan

Table 1 Data Management Plan

Data Description	Format	Spatial Resolution	Data of Acquisition	CRS
Luchtmeet Air Quality Data	Vector Layer (.geojson)	-	31-12-2024	EPSG:28992
Samen Meten	Vector Layer (.geojson)	-	31-12-2024	EPSG:28992
Administrative Boundaries	Vector Layer (.geojson)	-	31-12-2025 (The Netherlands)	EPSG:28992
			31-12-2025 (The Hague)	EPSG:28992
Schools	Vector Layer (.geojson)	-	31-12-2025 (The Hague)	EPSG:28992
Kontur Population Data	Vector Layer (.gpkg)	400m	31-12-2025 (The Hague)	EPSG:28992

Comparison between Official & Community Air Quality

